Lindblad Quantum Simulator – Quick User Guide

1. Overview

The Lindblad Quantum Simulator is a browser-based tool for studying quantum tunneling and decoherence in a Josephson-junction (superconducting qubit) system. It runs entirely in the browser: open `index.html` in a modern browser (Chrome, Firefox, Edge or Safari); no installation or internet connection is required.

2. Quick Start

1. Open `index.html` in your browser.
2. Set the physical parameters in the control panel (or pick a preset experiment).
3. Choose an initial state and a simulation method.
4. Click ▶ **Start Evolution** to run; use ⏸ **Pause** to stop temporarily and ↺ **Reset** to return to the initial state.
5. Watch the plots and result panels update in real time, then export what you need.

3. Setting Parameters

System parameters. Josephson energy EJ, charging energy EC, external flux Φ/Φ_o and bias current I define the potential and the energy levels. Hilbert space dimension sets the basis size (larger is more accurate but slower), and time step Δt sets the integration step.

Environment parameters. Temperature T, relaxation rate Γ_1 and dephasing rate $\Gamma\phi$ control decoherence. For noise studies, set the 1/f flux-noise amplitude, the low and high cutoff frequencies and the noise memory time. The number of MCWF trajectories controls how many quantum trajectories are averaged.

Recommended settings

The table lists each control with its unit, slider range and default value. Energies are entered as E/h in GHz and times in ns; internally the simulator converts them to SI units.

Parameter	Unit	Range	Default
Josephson energy EJ/h	GHz	5 – 30	15.0
Charging energy EC/h	GHz	0.1 – 2	0.8
External flux Φ/Φ_o	–	0 – 1	0.50
Bias current Ib/Ic	–	0 – 0.9	0.30
Hilbert space dimension N	–	8 – 64 (steps of 8)	32
Time step Δt	ns	0.001 – 0.02	0.005
Temperature T	mK	5 – 100	20

Parameter	Unit	Range	Default
Relaxation rate Γ_1	MHz	0 – 1000	0.1
Dephasing rate $\Gamma\phi$	MHz	0 – 500	0.05
MCWF trajectories	–	1 – 500	100
1/f noise amplitude $A\Phi$	$\mu\Phi_0/\sqrt{Hz}$	0.1 – 10	1.0
Noise cutoff fmin	Hz	0.1 – 10	1.0
Noise cutoff fmax	Hz	10 – 10000	1000
Noise memory time	µs	0.1 – 100	10.0

Model validity. The simulator treats the device in the transmon regime and flags results when EJ/EC falls below 20. With the default values EJ/EC = 15.0/0.8 = 18.75, so for transmon-regime studies raise EJ or lower EC (for example EJ = 20 GHz, EC = 0.5 GHz gives a ratio of 40).

Worked example: watching a superposition decohere

1. Click ↺ **Reset** and keep the default system parameters, or set EJ = 20 GHz and EC = 0.5 GHz to stay in the transmon regime.
2. Choose **Cat State (|0⟩+|1⟩)/√2** as the initial state.
3. Select **Pure Unitary** and click ▶ **Start Evolution**. On the **Bloch Sphere** the state vector stays on the surface and precesses, and purity remains 1.
4. Click ↺ **Reset**, choose **Lindblad (Density Matrix)** and **With Decoherence**, and set Γ_1 = 1 MHz and $\Gamma\phi$ = 0.5 MHz. Start again: the vector now shrinks toward the axis, the off-diagonal elements decay, and the system relaxes toward |0⟩.
5. Switch to **MCWF (Quantum Trajectories)** with 100 trajectories and compare **Single Trajectory** (random quantum jumps) with **Ensemble Average**, which should reproduce the Lindblad curve.
6. Use 📌 **Save Set B** before changing rates to compare two runs, and 📊 **CSV** or 📥 **Export Results (JSON)** to keep the data.

4. Initial State and Method

Initial state: ground state |0⟩, first excited state |1⟩, coherent state, cat state (|0⟩+|1⟩)/√2, or thermal state.

Simulation method: Unitary (wavefunction) for closed-system dynamics; MCWF (quantum trajectories) for stochastic open-system dynamics, shown as a single trajectory or an ensemble average; Lindblad (density matrix) for the full open-system master equation.

Dynamics mode: Pure Unitary, With Decoherence, or With 1/f Noise.

5. Reading the Results

The **Quantum State & Measurements** area shows the evolving state. Supporting panels report state properties, expectation values, decoherence rates, noise properties, tunneling statistics and potential

analysis. The **Bloch Sphere** visualises the qubit state, and the **Live Mathematics Panel** shows the equations being solved. **Real-time Performance** reports computation speed.

6. Validation and Analysis Tools

- **Validation Metrics** and **Comprehensive Benchmark (N=8→64)** check accuracy and scaling.
- **QuTiP vs JavaScript Validation** compares results with the QuTiP reference library.
- **1/f Noise PSD Analysis** shows the power spectral density of the generated noise.
- **Extended Validation Grid** and **Experimental Validation** compare against wider parameter sets and published data.
- **Parameter Comparison:** run one configuration, click 📌 **Save Set B**, change parameters, and use 🔄 **Swap A↔B** to compare the two.

7. Learning Features

What is Quantum Tunneling?, **Key Concepts** and **Historical Context** give background. **Preset Experiments** load ready-made scenarios. The **Knowledge Quiz** tests understanding (🔄 **Restart** to retry, 📋 **Show Answers** to reveal solutions). The **Educator Dashboard** supports classroom use.

8. Exporting and Sharing

- 📥 **Export Results (JSON)** saves the full simulation output.
- 📷 **PNG** saves the current figure; 📊 **CSV** saves the data for spreadsheets.
- 🎓 **Student Log** records a student's session.
- 🔗 **URL** creates a link that reproduces the current parameter set.
- 👁 **Contrast** switches to a high-contrast display.

9. Tips

- Start with a small Hilbert space and few trajectories, then increase them once the setup looks right.
- If the simulation slows down, reduce the Hilbert space dimension or the number of trajectories, or increase Δt carefully.
- Use ↺ **Reset** after changing the initial state or method so that the run starts cleanly.